



Applied Statistical Analysis of COVID-19 Patterns Across EU/EEA Countries

Group 1 – Final Project Presentation
MSDS 401: Applied Statistics with R

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Dataset Overview & Research Objectives

Dataset Overview

- Daily COVID-19 cases and deaths reported by ECDC for all EU/EEA countries, 2020–2022
- Covers 30 countries, over 1000 daily observations per country
- Standardized using population data (2020)

Country Selection

Six countries selected for deeper analysis:

Northern Europe	Western Europe		Southern Europe	Eastern Europe	
Finland	France	Germany	Italy	Czechia	Lithuania

Selection represents all major European regions

Research Purpose

- Understand transmission patterns
- Compare mortality outcomes
- Test statistical differences
- Build predictive models to explain cross-country variation

Data Preparation

Standardized Epidemiological Measures:

Incidence rate
(cases per 100,000)

Fatality rate
(deaths per 100,000)

Cleaning & Formatting

Converted date formats and simplified variable names

Kept negative values (these reflect official reporting corrections)

Regional Assignment

Used UN M49 regions, with Estonia, Latvia, and Lithuania reassigned from Northern to Eastern Europe for analytical consistency

Rolling Averages

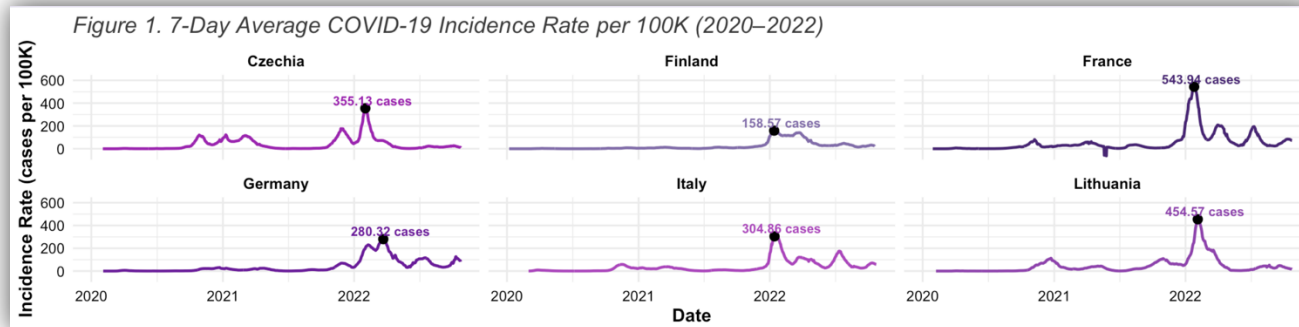
Applied 7-day smoothing to reduce noise from weekend lags and reporting delays.

Selection of Focus Countries

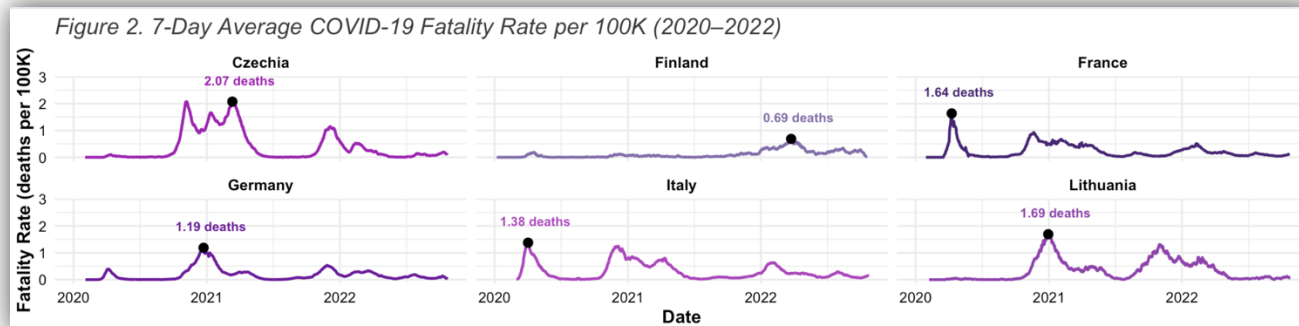
Six countries chosen to represent all regions: Finland (North), France & Germany (West), Italy (South), Czechia & Lithuania (East)

Preliminary Exploratory Data Analysis (EDA)

- 7-day rolling averages were used to smooth reporting noise and highlight true epidemic patterns.
- Countries experienced different timing and intensity of epidemic waves, visible in both incidence and fatality curves.
- Late 2021–2022 shows strong surges across most countries (Omicron period), with a clear lag between cases and deaths.
- Finland shows the mildest trends, while France, Czechia, Italy, and Lithuania exhibit sharper, higher peaks.



The six countries display sharply different transmission patterns. France, Czechia, Italy, and Lithuania show high, intense peaks, especially during Omicron, while Finland remains consistently low.



Fatality peaks lag incidence by 2–4 weeks. Czechia and Lithuania show the highest mortality peaks; Finland again remains lowest. A strong case–death decoupling appears in late pandemic waves.

Literature Review

Structural factors matter

Population size, density, age structure, and economic capacity shape national COVID-19 outcomes (OECD, 2021).

Vaccination reduces fatality

Vaccination campaigns lowered mortality by up to 59% in the WHO European Region (Meslé et al., 2024).

Wave differences across Europe

Variants, policy measures, and healthcare capacity drove differences in wave timing and severity (ECDC, 2022; WHO, 2022).

Correlation studies

Prior research found incidence–fatality correlations vary by country due to vaccination, testing, and demographics (Iqbal et al., 2022).

Policy & behavior influence outcomes

Public compliance, mask-wearing, and mobility patterns shaped infection levels (Bontempi, 2021; Tsiang, 2021).

Hypothesis Testing: Germany vs Italy

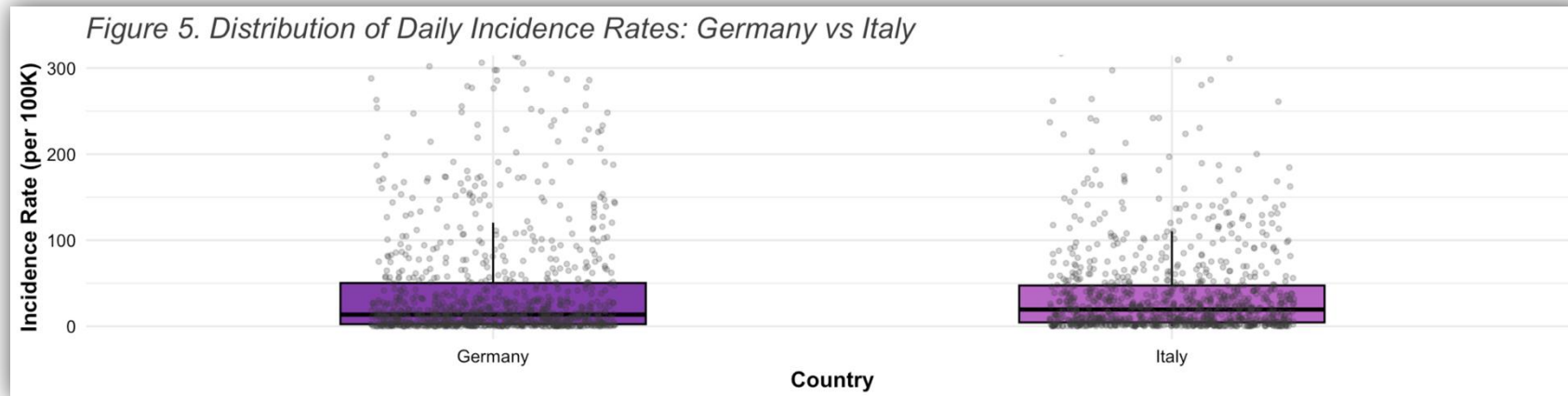
Objective: Determine whether Germany and Italy differ in their mean daily COVID-19 incidence rates (per 100K).

Methodology

- Independent two-sample t-test (pooled variance).
- Standardized incidence rates used for comparability.
- Significance level $\alpha = 0.05$.
- Wilcoxon rank-sum test added as a robustness check.

Assumptions checked:

- Independence (separate populations)
- Large samples → normality reasonable
- Levene's test $p = 0.105$ → equal variances assumed
- Wilcoxon used due to skewed distributions (nonparametric alternative)

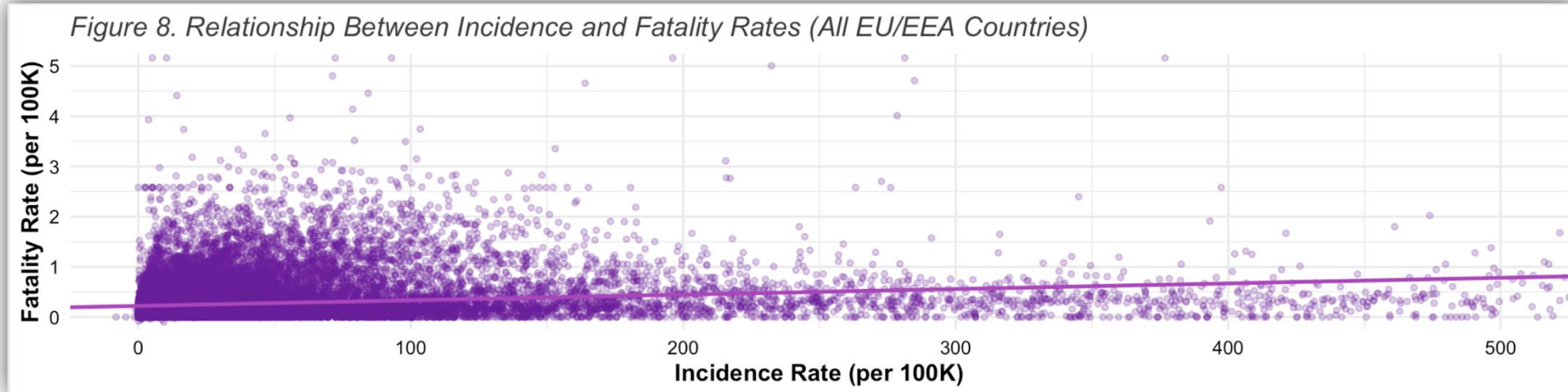


t-Statistic	p-Value	95% Confidence Interval	Decision
0.887	0.375	-3.10 to 8.22	Fail to Reject H_0

*Wilcoxon rank-sum test also indicates a small but significant difference between the distributions ($p = 0.011$).

Germany and Italy show similar mean incidence levels, and the observed difference is likely due to random variation rather than a true population difference.

Correlation Between Incidence and Fatality Rates



- The Pearson correlation across all EU/EEA countries is $r = 0.1126$, statistically significant but weak.
- Higher incidence days are associated with slightly higher fatality rates, but the relationship is not strong.
- Mortality outcomes are influenced by many external factors (vaccination, age structure, healthcare capacity).
- Stronger country-level relationships appear in Finland, Lithuania, and Czechia; weaker in Germany, Italy, and France.
- Highlights the need to examine correlation within each country, not only at the pooled European level.

Correlation (r)	p-Value	95% CI (Lower)	95% CI (Upper)	Interpretation
0.1126	< 0.001	0.1011	0.1241	Weak positive correlation

Correlation Patterns Across the Six Selected Countries

- Correlation strength varies substantially across countries.
- Finland shows the strongest association between incidence and fatality, indicating closely linked wave patterns.
- Czechia and Lithuania show moderate positive correlations.
- Germany, Italy, and France show only weak correlations, reflecting the decoupling of cases and deaths in later waves.
- Results highlight that country-level factors (healthcare strain, vaccination timing, demographics) strongly influence the link between transmission and mortality.

Country	r	Interpretation
Finland	0.8395	Strong positive
Czechia	0.4953	Moderate positive
Lithuania	0.4415	Moderate positive
Germany	0.1937	Weak positive
Italy	0.1450	Weak positive
France	0.0789	Weak positive

Prediction Analysis (Regression Summary)

Assessing How Country-Level Structural Factors Predict Total COVID-19 Case Counts

- We built a multiple linear regression model using 20 EU/EEA countries.
- Predictors included: Population size, Population density, and GDP per capita (PPS).
- The response variable was total cumulative COVID-19 cases (2020–2022).
- Objective: Determine which structural factors best explain cross-country variation in total case burden.
- Results show population size is the only statistically significant predictor.
- Density and GDP per capita do not meaningfully improve the model once population is included.

Total Cases = β_0 + β_1 (Population) + β_2 (Population Density) + β_3 (GDP per Capita)

Significant Predictor	Non-Significant Predictors	R ²	Adjusted R ²	Model Significance (F-test)
Population size (p < 0.001)	Density, GDP per capita	0.8982	0.8791	p < 0.001

Prediction Analysis (Model Validation)

Evaluating How Well the Regression Model Performs on New Countries

- Test whether the regression model generalizes beyond the 20 countries used in training.
- Predict cumulative COVID-19 cases for: Netherlands and Luxembourg
- Compare Predicted vs Actual case totals.

➤ Netherlands:

✓ Predicted value is reasonably close → model generalizes well to large, structurally similar countries.

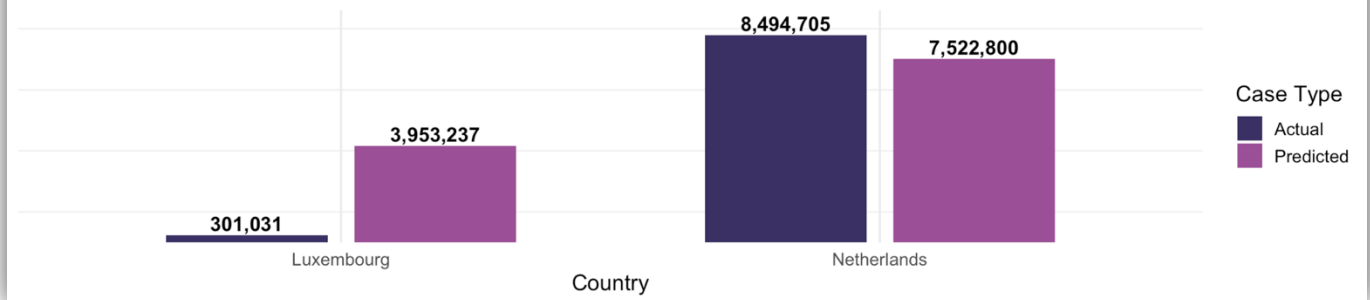
➤ Luxembourg:

✗ Severe overestimation → model fails for microstates with small populations and extremely high GDP.

➤ Key Insight:

The model performs only within the structural range of the training data and cannot reliably extrapolate to atypical countries.

Figure 10. Actual vs Predicted Total COVID-19 Cases



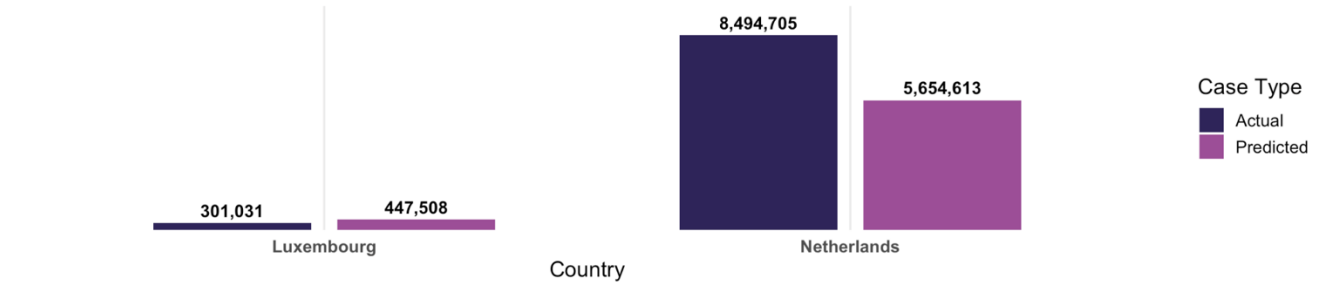
Metric	Netherlands	Luxembourg
Residual	+971,865	-3,649,000

Improved Model (Log-Transformed Regression)

Fixing Scaling Problems & Improving Prediction Stability

Original Model	Improved Log-Transformed Model
- Predicted 3,952,000 cases for Luxembourg (true = 301,031). - Large errors because the model assumed proportional linear growth. - High GDP and extreme density caused unrealistic predictions. - Struggled with microstates and structural outliers. Total Cases = $\beta_0 + \beta_1(\text{Population}) + \beta_2(\text{Population Density}) + \beta_3(\text{GDP per Capita})$	- Predictions become much closer to reality, especially for Luxembourg. - Relationships scale more naturally across countries of different sizes. - Reduces the influence of extreme GDP/density. - Still shows under-/overestimation → meaning structural variables alone are not enough. $\log(\text{Total Cases}) = \beta_0 + \beta_1 \cdot \log(\text{Population}) + \beta_2(\text{Population Density}) + \beta_3(\text{GDP per Capita})$

Figure 11. Improved Model: Actual vs Predicted Total COVID-19 Cases



Metric	Netherlands	Luxembourg
Residual	2,840,092	-146,477

Key Findings & Limitations

Key Findings

- ✓ **Population size** was the strongest predictor of total cases ($R^2 \approx 0.90$).
- ✓ **Incidence–fatality correlation is weak overall** ($r = 0.1126$), but varies strongly by country (e.g., Finland $r = 0.84$).
- ✓ **Finland shows the mildest epidemic**, while France, Czechia, Italy, and Lithuania show sharper peaks.
- ✓ **Germany vs Italy**: No significant difference in mean incidence ($p = 0.375$).
- ✓ **Improved log model** reduces extreme prediction errors, especially for Luxembourg.

Limitations

- ! **Structural variables alone are not sufficient** — missing vaccination, age, variant, healthcare capacity.
- ! **Luxembourg-like microstates break linear models** due to extreme GDP and high density.
- ! **Reporting corrections (negative values)** may introduce noise but had to be retained for cumulative totals.
- ! **Cross-country data inconsistencies** (different testing strategies, reporting delays).
- ! **Models assume linearity**, which does not reflect real epidemic behavior.

Future Research Directions

- Include vaccination coverage, variant circulation, and age structure.
- Model cases per 100K rather than total cases.
- Explore machine learning models for improved prediction.
- Add policy stringency indexes and testing intensity.
- Consider Bayesian or hierarchical models to capture country-level differences.

References

- Bontempi, E. 2021. "The Europe Second Wave of COVID-19 Infection and the Italy 'Strange' Situation." Environmental Research 193.
- ECDC. 2022. COVID-19 Surveillance Report: EU/EEA Overview. European Centre for Disease Prevention and Control. <https://www.ecdc.europa.eu>.
- Iqbal, H., M. Umair, S. A. Rizvi, and M. A. Cheema. 2022. "A Correlation Study of COVID-19 in Europe Considering Different Vaccines, Age Groups and Variants Including Delta and Omicron." BioMed 2 (2): 133–169. <https://doi.org/10.3390/biomed2020015>.
- Meslé, Margaux M. I., Jeremy Brown, Piers Mook, Mark A. Katz, José Hagan, Roberta Pastore, Bernhard Benka, et al. 2024. "Estimated Number of Lives Directly Saved by COVID-19 Vaccination Programmes in the WHO European Region from December 2020 to March 2023: A Retrospective Surveillance Study." The Lancet Respiratory Medicine 12 (9).
- OECD. 2021. Health at a Glance: Europe 2021 – State of Health in the EU Cycle. Organisation for Economic Co-operation and Development. <https://www.oecd.org>.
- Papanikos, Gregory T. 2022. "The Demographics of COVID-19 in the European Union." In Essays on COVID-19 Research, 217–228. Athens: AITNER.
- Tsiang, J. T. 2021. "COVID-19 Case and Death Rate Comparisons Between 5G mmWave and Non-5G mmWave Jurisdictions in the United States." Medical Research Archives 9 (11). <https://esmed.org/MRA/mra/article/view/2371>.
- WHO. 2022. WHO Regional Office for Europe—COVID-19 Dashboard and Epidemiological Updates. World Health Organization. <https://www.who.int>.

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